

PEMS03

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	2tta-ols	2tta-staff
MAE	3	19.05 ± 2.09	18.09 ± 1.08	18.49 ± 1.05	18.56 ± 1.05	19.21 ± 0.93	19.35 ± 0.97	19.60 ± 0.99	19.40 ± 0.99	19.71 ± 0.10	19.48 ± 0.89	19.89 ± 0.10	19.67 ± 0.30	19.31 ± 1.97	19.72 ± 0.50	19.53 ± 12.10	19.75 ± 0.10	19.21 ± 0.86	19.86 ± 0.34	19.34 ± 0.03	
	6	19.27 ± 2.09	18.97 ± 1.08	18.86 ± 1.17	18.17 ± 1.05	19.53 ± 0.98	19.98 ± 0.10	19.24 ± 1.06	19.06 ± 1.03	19.35 ± 1.08	19.08 ± 1.06	19.48 ± 1.16	19.61 ± 1.29	19.65 ± 1.03	19.38 ± 1.57	19.58 ± 12.95	19.40 ± 1.12	19.52 ± 0.87	19.70 ± 0.03		
	12	20.27 ± 2.96	21.96 ± 1.93	19.93 ± 1.26	19.69 ± 1.26	20.46 ± 1.32	19.32 ± 1.53	19.53 ± 1.40	19.70 ± 1.70	19.38 ± 1.84	19.84 ± 1.36	19.28 ± 1.30	21.78 ± 1.58	19.15 ± 14.94	19.81 ± 1.81	19.21 ± 1.08	19.80 ± 1.31	19.31 ± 0.03			
	Avg	19.45 ± 2.17	21.17 ± 1.91	19.01 ± 1.34	19.66 ± 1.10	19.34 ± 1.07	19.17 ± 1.04	19.47 ± 1.20	19.62 ± 1.18	19.81 ± 1.30	19.08 ± 1.50	19.60 ± 13.17	19.53 ± 1.33	19.61 ± 0.87	19.72 ± 0.04						
RMSE	3	28.63 ± 26.95	32.98 ± 17.53	32.20 ± 15.62	36.13 ± 15.79	36.13 ± 17.13	36.40 ± 17.46	36.63 ± 16.23	38.72 ± 19.43	36.24 ± 17.22	34.96 ± 14.06	34.06 ± 0.04									
	6	29.11 ± 28.76	33.57 ± 18.76	33.25 ± 16.97	37.46 ± 17.15	37.49 ± 18.45	37.71 ± 19.51	36.23 ± 17.63	38.87 ± 20.87	37.54 ± 19.17	36.29 ± 14.82	36.82 ± 0.03									
	12	31.13 ± 22.67	35.30 ± 21.76	35.80 ± 19.64	39.99 ± 19.74	40.16 ± 21.14	40.49 ± 23.73	39.16 ± 20.42	39.61 ± 24.35	40.25 ± 23.36	38.72 ± 16.04	38.03 ± 0.03									
	Avg	29.45 ± 29.11	33.80 ± 19.10	33.53 ± 17.16	37.62 ± 17.32	37.68 ± 18.65	37.94 ± 19.89	37.18 ± 17.84	38.90 ± 21.27	37.76 ± 19.54	36.42 ± 14.84	36.84 ± 0.03									
MAPE (%)	3	74.40 ± 278.76	36.64 ± 12.27	58.98 ± 40.11	29.01 ± 29.40	30.85 ± 33.49	32.05 ± 34.80	26.12 ± 171.05	51.26 ± 35.66	30.16 ± 29.48	27.00 ± 26.39	26.39 ± 0.29									
	6	74.81 ± 270.39	37.04 ± 23.27	57.79 ± 31.11	29.77 ± 30.08	31.36 ± 34.15	32.68 ± 36.55	25.32 ± 132.08	61.98 ± 37.86	31.16 ± 31.56	27.88 ± 26.91	26.91 ± 0.25									
	12	76.89 ± 274.42	40.28 ± 26.05	57.02 ± 23.86	32.06 ± 32.36	33.79 ± 36.24	35.23 ± 41.76	20.92 ± 234.91	63.43 ± 23.56	33.78 ± 37.14	30.06 ± 28.03	28.03 ± 0.23									
	Avg	75.13 ± 270.85	37.72 ± 23.62	58.10 ± 31.44	30.11 ± 30.41	31.78 ± 34.40	33.10 ± 37.22	26.77 ± 132.42	62.05 ± 38.56	31.48 ± 32.27	28.12 ± 27.01	27.01 ± 0.25									

PEMS04

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinea	2tta-ol	2tta-sta
MAE	3	99.44 ± 1.06	106.47 ± 1.11	111.23 ± 1.18	118.43 ± 1.35	151.51 ± 1.56	162.12 ± 1.65	174.05 ± 1.70	170.05 ± 1.66	191.09 ± 1.46	180.46 ± 1.26	185.26 ± 1.31	191.01 ± 1.26	206.12 ± 1.06	196.85 ± 1.08	183.31 ± 1.06	165.05 ± 1.03	139.79 ± 1.13	139.91 ± 1.03	133.06 ± 0.98	
	6	99.58 ± 1.08	108.04 ± 1.11	111.37 ± 1.19	125.38 ± 1.57	135.31 ± 1.68	161.81 ± 1.69	169.91 ± 1.68	168.85 ± 1.20	203.30 ± 1.47	179.79 ± 1.28	184.45 ± 1.39	189.69 ± 1.28	207.31 ± 1.07	197.07 ± 1.09	194.49 ± 1.72	172.21 ± 1.40	140.87 ± 1.25	152.23 ± 1.03	133.78 ± 0.98	
	12	100.27 ± 1.20	105.56 ± 1.11	111.81 ± 1.22	125.00 ± 1.35	152.52 ± 1.59	197.97 ± 1.09	194.07 ± 1.40	193.35 ± 1.09	227.79 ± 1.20	204.43 ± 1.31	219.95 ± 1.37	207.04 ± 1.25	209.97 ± 1.07	197.45 ± 1.22	223.35 ± 1.09	198.82 ± 1.43	163.67 ± 1.27	178.80 ± 1.04	144.95 ± 0.98	
	Avg	99.70 ± 1.08	108.39 ± 1.11	111.44 ± 1.19	119.67 ± 1.35	135.47 ± 1.57	157.58 ± 1.70	174.01 ± 1.71	171.12 ± 1.67	170.07 ± 1.20	205.54 ± 1.48	181.12 ± 1.28	188.80 ± 1.39	189.42 ± 1.28	207.58 ± 1.07	197.22 ± 1.09	198.82 ± 1.74	174.47 ± 1.41	141.20 ± 1.25	154.41 ± 1.03	133.81 ± 0.98
RMSE	3	120.58 ± 1.23	129.98 ± 1.69	139.79 ± 1.28	168.45 ± 1.68	215.35 ± 1.75	248.84 ± 1.24	248.85 ± 1.24	248.86 ± 1.24	297.72 ± 1.26	266.04 ± 1.38	281.57 ± 1.10	277.27 ± 1.25	315.35 ± 1.05	313.72 ± 1.28	288.28 ± 1.25	252.26 ± 1.51	215.59 ± 1.23	238.11 ± 1.03	211.95 ± 0.98	
	6	120.77 ± 1.23	129.76 ± 1.69	139.75 ± 1.29	168.62 ± 1.68	217.60 ± 1.77	268.87 ± 1.26	277.05 ± 1.27	277.06 ± 1.31	317.75 ± 1.28	288.04 ± 1.41	311.20 ± 1.19	316.27 ± 1.26	353.93 ± 1.30	350.19 ± 1.27	344.40 ± 1.54	304.05 ± 1.25	254.40 ± 1.03	273.32 ± 0.98		
	12	121.72 ± 1.52	129.63 ± 1.69	139.73 ± 1.34	169.82 ± 1.69	222.22 ± 1.72	307.99 ± 1.30	311.28 ± 1.31	311.37 ± 1.36	366.04 ± 1.52	327.77 ± 1.46	368.82 ± 1.16	364.70 ± 1.28	422.22 ± 1.05	424.07 ± 1.34	374.75 ± 1.31	319.92 ± 1.59	259.49 ± 1.29	296.61 ± 1.03	255.39 ± 0.98	
	Avg	120.93 ± 1.29	129.27 ± 1.69	139.76 ± 1.30	168.89 ± 1.68	217.99 ± 1.77	274.14 ± 1.27	277.35 ± 1.27	277.37 ± 1.32	312.12 ± 1.28	288.55 ± 1.41	311.75 ± 1.18	316.16 ± 1.26	356.99 ± 1.05	354.00 ± 1.30	340.69 ± 1.27	304.80 ± 1.54	254.57 ± 1.25	275.65 ± 1.03	253.34 ± 0.98	
MAPE (%)	3	494.29 ± 1.62	494.49 ± 1.68	498.08 ± 1.43	498.23 ± 1.84	498.88 ± 1.21	498.86 ± 1.20	497.28 ± 1.19	497.72 ± 1.21	495.56 ± 1.24	494.61 ± 1.40	490.84 ± 1.70	490.46 ± 1.71	491.86 ± 1.43	493.91 ± 1.33	493.41 ± 1.22	492.32 ± 1.34	494.47 ± 1.18	498.12 ± 1.07	497.51 ± 0.98	
	6	494.94 ± 1.62	493.03 ± 1.68	498.34 ± 1.43	493.92 ± 1.84	494.71 ± 1.22	492.96 ± 1.21	491.20 ± 1.20	490.73 ± 1.22	496.80 ± 1.25	495.14 ± 1.41	491.81 ± 1.69	490.02 ± 1.62	492.96 ± 1.44	494.29 ± 1.34	494.86 ± 1.23	493.20 ± 1.35	495.75 ± 1.19	499.12 ± 1.07	497.92 ± 0.98	
	12	495.68 ± 1.62	497.12 ± 1.69	499.16 ± 1.46	496.23 ± 1.84	494.46 ± 1.25	495.99 ± 1.23	493.53 ± 1.23	493.51 ± 1.25	495.29 ± 1.29	499.63 ± 1.57	497.74 ± 1.45	495.09 ± 1.67	497.94 ± 1.25	495.99 ± 1.45	498.59 ± 1.38	495.55 ± 1.39	499.86 ± 1.21	499.55 ± 1.08	498.94 ± 0.98	
	Avg	494.79 ± 1.62	493.53 ± 1.68	498.45 ± 1.44	494.28 ± 1.84	494.72 ± 1.23	493.35 ± 1.21	491.49 ± 1.21	491.09 ± 1.22	497.75 ± 1.27	497.09 ± 1.25	495.55 ± 1.42	495.17 ± 1.69	493.35 ± 1.63	494.51 ± 1.35	495.33 ± 1.23	493.52 ± 1.36	496.38 ± 1.19	499.40 ± 1.08	498.03 ± 0.98	

PEMS05

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	2tta-ols	2tta-staff
MAE	3	8.41 ± 0.8	8.02 ± 0.8	8.24 ± 0.8	8.52 ± 0.2	10.46 ± 0.8	8.02 ± 0.7	6.69 ± 0.7	7.74 ± 0.7	7.87 ± 0.8	8.17 ± 0.7	9.94 ± 0.1	11.48 ± 0.3	23.19 ± 0.5	8.02 ± 0.1	10.61 ± 0.4	9.10 ± 0.7	7.98 ± 0.8	8.13 ± 0.7	7.22 ± 0.6	6.79 ± 0.04
	6	8.64 ± 0.8	8.66 ± 0.8	8.58 ± 0.9	9.02 ± 0.2	10.77 ± 0.8	8.47 ± 0.8	8.07 ± 0.8	8.18 ± 0.8	8.28 ± 0.8	8.62 ± 0.8	8.36 ± 0.1	11.92 ± 0.3	23.27 ± 0.5	8.47 ± 0.1	10.94 ± 0.4	9.46 ± 0.8	8.44 ± 0.8	8.81 ± 0.7	7.66 ± 0.6	6.99 ± 0.04
	12	9.39 ± 0.9	9.97 ± 0.9	9.43 ± 0.1	10.32 ± 0.1	11.62 ± 1.9	9.41 ± 0.8	8.85 ± 0.9	9.04 ± 0.9	9.14 ± 0.9	9.57 ± 0.9	9.25 ± 0.1	12.93 ± 0.3	24.37 ± 0.5	9.41 ± 0.1	13.67 ± 0.4	10.26 ± 0.9	9.38 ± 0.1	10.30 ± 0.8	8.50 ± 0.7	7.33 ± 0.06
	Avg	8.75 ± 0.8	8.77 ± 0.8	8.68 ± 0.9	9.19 ± 0.2	10.89 ± 0.8	8.55 ± 0.8	8.13 ± 0.8	8.25 ± 0.8	8.36 ± 0.8	8.70 ± 0.8	8.44 ± 0.1	12.02 ± 0.3	23.50 ± 0.5	8.55 ± 0.1	10.02 ± 0.4	9.54 ± 0.8	8.52 ± 0.8	8.95 ± 0.7	7.72 ± 0.7	7.00 ± 0.05
RMSE	3	12.50 ± 0.2	12.08 ± 0.1	12.35 ± 0.2	12.58 ± 0.5	25.52 ± 1.1	11.98 ± 0.1	11.52 ± 0.1	11.62 ± 0.1	11.80 ± 0.2	12.17 ± 0.1	11.89 ± 0.5	15.88 ± 0.3	31.12 ± 0.1	11.98 ± 0.1	17.90 ± 0.3	13.15 ± 0.1	11.96 ± 0.2	12.27 ± 0.1	11.01 ± 0.1	10.21 ± 0.08
	6	13.03 ± 0.3	13.25 ± 0.3	13.06 ± 0.3	13.49 ± 0.6	26.09 ± 1.2	12.86 ± 0.2	12.21 ± 0.2	12.48 ± 0.2	12.58 ± 0.3	13.07 ± 0.2	12.70 ± 0.6	16.65 ± 0.3	31.67 ± 0.1	12.86 ± 0.3	18.44 ± 0.3	13.75 ± 0.2	12.87 ± 0.3	13.52 ± 0.1	11.86 ± 0.1	10.56 ± 0.08
	12	14.50 ± 0.5	15.55 ± 0.4	14.68 ± 0.5	15.69 ± 0.7	27.46 ± 1.4	14.65 ± 0.3	13.51 ± 0.4	14.07 ± 0.4	14.16 ± 0.4	14.87 ± 0.4	14.34 ± 0.8	18.24 ± 0.3	33.65 ± 0.1	14.65 ± 0.9	19.65 ± 0.3	15.02 ± 0.4	14.63 ± 0.6	16.09 ± 0.3	13.37 ± 0.1	11.12 ± 0.10
	Avg	13.22 ± 0.3	13.42 ± 0.3	13.22 ± 0.3	13.75 ± 0.6	26.25 ± 1.3	13.01 ± 0.2	12.29 ± 0.2	12.58 ± 0.2	12.71 ± 0.3	13.21 ± 0.2	12.84 ± 0.6	16.78 ± 0.3	31.94 ± 0.1	13.01 ± 0.8	18.57 ± 0.3	13.86 ± 0.3	13.00 ± 0.3	13.74 ± 0.1	11.94 ± 0.1	10.57 ± 0.09
MAPE (%)	3	33.99 ± 0.1	31.42 ± 0.3	33.07 ± 0.1	31.75 ± 0.2	67.97 ± 0.3	33.01 ± 0.0	30.28 ± 0.9	29.57 ± 0.1	31.84 ± 0.4	34.22 ± 0.3	32.55 ± 0.1	41.33 ± 0.2	92.82 ± 0.3	33.01 ± 0.5	35.51 ± 0.3	32.45 ± 0.1	31.31 ± 0.3	32.31 ± 0.2	27.47 ± 0.48	27.57 ± 0.48
	6	34.37 ± 0.3	33.51 ± 0.3	33.88 ± 0.2	32.97 ± 0.2	68.59 ± 0.2	32.06 ± 0.1	31.34 ± 0.0	30.32 ± 0.2	32.67 ± 0.3	35.22 ± 0.3	33.80 ± 0.1	41.86 ± 0.2	92.97 ± 0.3	34.06 ± 0.6	36.57 ± 0.3	33.80 ± 0.2	32.30 ± 0.4	34.60 ± 0.2	28.42 ± 0.47	28.03 ± 0.47
	12	36.05 ± 0.3	38.05 ± 0.3	38.74 ± 0.6	36.09 ± 0.2	72.23 ± 0.5	35.85 ± 0.3	33.80 ± 0.2	32.43 ± 0.4	34.98 ± 0.3	37.80 ± 0.2	37.21 ± 0.6	46.41 ± 0.2	99.04 ± 0.3	36.85 ± 0.9	39.26 ± 0.3	37.37 ± 0.4	34.83 ± 0.1	40.50 ± 0.4	30.44 ± 0.4	28.93 ± 0.55
	Avg	34.68 ± 0.3	33.93 ± 0.3	36.32 ± 0.3	33.35 ± 0.2	69.51 ± 0.3	33.40 ± 0.1	31.58 ± 0.0	30.57 ± 0.2	32.95 ± 0.3	35.53 ± 0.3	34.21 ± 0.2	42.72 ± 0.2	93.85 ± 0.3	34.40 ± 0.6	36.91 ± 0.3	34.23 ± 0.2	32.61 ± 0.3	35.31 ± 0.2	28.60 ± 0.4	28.11 ± 0.52

PEMS06

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	2tta-ols	2tta-staff
MAE	3	14.26 ± 2.04	10.47 ± 0.56	11.17 ± 0.11	65.79 ± 10.13	09.92 ± 0.98	09.84 ± 0.99	09.98 ± 0.10	10.01 ± 3.03	09.39 ± 0.45	05.55 ± 35.96	8.013 ± 0.49	09.20 ± 32.08	01.036 ± 0.076	09.26 ± 0.80	06.64 ± 0.05					
	6	14.52 ± 3.11	11.33 ± 0.60	11.87 ± 0.11	66.36 ± 10.76	10.52 ± 0.46	10.62 ± 0.25	10.99 ± 0.56	15.61 ± 35.52	8.076 ± 0.49	09.25 ± 42.66	01.098 ± 0.174	09.91 ± 0.80	09.93 ± 0.05							
	12	15.57 ± 23.10	17.07 ± 0.34	13.48 ± 0.66	66.88 ± 12.06	11.72 ± 0.16	11.87 ± 0.37	22.30 ± 0.78	23.40 ± 12.06	09.72 ± 33.91	02.32 ± 0.38	01.110 ± 0.41	09.41 ± 0.07								
	Avg	14.70 ± 3.14	11.47 ± 0.61	12.04 ± 0.66	66.26 ± 10.86	10.61 ± 0.55	10.71 ± 0.26	11.10 ± 0.58	15.82 ± 35.05	8.086 ± 0.49	09.34 ± 32.78	01.110 ± 0.193	09.98 ± 0.80	09.94 ± 0.06							
RMSE	3	20.17 ± 36.02	24.68 ± 0.68	31.19 ± 15.43	15.16 ± 0.49	15.17 ± 0.88	15.71 ± 20.88	19.79 ± 115.43	68.42 ± 47.65	05.64 ± 0.65	06.51 ± 0.41	16 ± 13.05	0.07								
	6	20.78 ± 37.58	25.44 ± 0.81	32.22 ± 16.66	16.34 ± 0.61	16.40 ± 0.98	16.92 ± 22.76	19.64 ± 106.66	68.68 ± 48.59	06.80 ± 0.83	06.38 ± 0.53	16.59 ± 13.59	0.08								
	12	22.72 ± 20.49	27.18 ± 0.85	33.64 ± 18.95	18.48 ± 0.83	18.37 ± 0.65	21.80 ± 19.28	26.57 ± 37.97	98.95 ± 69.49	20.58 ± 0.19	01.16 ± 0.21	01.83 ± 0.77	14.45 ± 0.11								
	Avg	21.06 ± 37.75	25.62 ± 0.83	32.15 ± 16.79	16.45 ± 0.62	16.29 ± 0.52	19.99 ± 17.08	23.08 ± 19.15	106.79 ± 68.78	48.77 ± 0.69	08.56 ± 0.54	15.49 ± 13.60	0.09								
MAPE (%)	3	58.19 ± 222.49	28.46 ± 24.59	15.82 ± 23.24	21.21 ± 21.67	21.90 ± 27.02	25.97 ± 36.18	167.56 ± 53.24	50.16 ± 33.48	24.85 ± 23.70	19.77 ± 19.27	0.34									
	6	58.57 ± 223.77	28.85 ± 25.64	15.88 ± 24.13	21.92 ± 22.56	22.51 ± 27.78	26.48 ± 37.23	169.33 ± 54.13	50.11 ± 34.58	25.84 ± 25.54	20.54 ± 19.58	0.33									
	12	60.22 ± 227.09	30.64 ± 28.30	15.47 ± 26.52	23.35 ± 24.51	24.22 ± 29.63	28.19 ± 39.35	171.72 ± 176.52	50.99 ± 37.39	28.02 ± 30.02	22.12 ± 20.24	0.25									
	Avg	58.90 ± 224.15	29.16 ± 25.94	15.85 ± 24.41	22.03 ± 22.75	22.72 ± 27.97	26.68 ± 37.39	169.42 ± 54.41	50.35 ± 34.90	26.03 ± 26.05	20.67 ± 19.63	0.31									

PEMS07

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	2tta-ols	2tta-staff
MAE	3	63.07 ± 5.08	62.15 ± 5.16	66.92 ± 10.98	64.53 ± 13.57	63.98 ± 13.81	66.10 ± 13.89	64.93 ± 63.42	64.53 ± 65.06	61.62 ± 14.69	65.01 ± 62.90	62.01 ± 0.09									
	6	63.36 ± 5.67	66.34 ± 17.81	61.13 ± 15.86	64.75 ± 65.39	65.14 ± 67.34	65.25 ± 65.87	67.40 ± 65.86	65.91 ± 63.50	66.03 ± 66.89	64.30 ± 62.77	0.10									
	12	64.80 ± 5.04	66.74 ± 20.20	61.26 ± 18.48	66.82 ± 67.92	67.54 ± 69.63	67.72 ± 67.77	68.50 ± 67.48	68.38 ± 67.74	68.74 ± 60.97	66.76 ± 63.90	0.15									
	Avg	63.62 ± 5.71	66.38 ± 18.10	61.08 ± 16.04	64.84 ± 65.52	65.27 ± 67.47	65.38 ± 66.00	68.98 ± 66.04	66.23 ± 63.95	66.23 ± 67.26	64.41 ± 62.76	0.11									
RMSE	3	89.39 ± 64.87	70.14 ± 27.20	51.60 ± 23.85	62.69 ± 23.21	62.96 ± 25.78	63.10 ± 24.22	62.01 ± 23.85	65.05 ± 24.19	64.09 ± 65.12	62.06 ± 20.92	0.16									
	6	89.94 ± 67.95	70.04 ± 28.79	52.70 ± 26.35	64.82 ± 25.69	65.39 ± 68.03	65.54 ± 66.02	69.43 ± 66.35	66.46 ± 67.19	66.49 ± 68.68	64.44 ± 62.37	0.21									
	12	92.24 ± 54.29	70.68 ± 32.78	54.33 ± 30.92	68.33 ± 29.90	69.44 ± 61.97	69.65 ± 69.15	68.34 ± 63.02	63.31 ± 64.26	61.02 ± 65.79	68.37 ± 64.44	0.30									
	Avg	90.32 ± 68.45	70.02 ± 29.22	52.68 ± 26.60	64.91 ± 25.84	65.52 ± 68.21	65.68 ± 66.13	71.90 ± 66.60	66.89 ± 67.97	66.76 ± 69.20	64.54 ± 62.33	0.21									
MAPE (%)	3	227.57 ± 93.75	65.74 ± 20.66	62.64 ± 18.95	67.91 ± 68.00	67.45 ± 60.72	67.57 ± 60.65	65.23 ± 68.95	61.64 ± 69.89	69.54 ± 68.08	65.85 ± 65.16	0.19									
	6	227.77 ± 90.21	66.28 ± 21.45	61.89 ± 20.03	68.78 ± 68.98	68.49 ± 61.65	68.70 ± 61.81	67.38 ± 68.03	62.66 ± 62.65	60.57 ± 69.76	66.98 ± 65.67	0.17									
	12	229.22 ± 90.00	67.77 ± 23.99	60.94 ± 22.56	60.88 ± 61.24	60.84 ± 63.76	61.32 ± 64.48	63.85 ± 62.56	66.08 ± 69.11	63.04 ± 64.21	69.42 ± 66.71	0.24									
	Avg	228.08 ± 90.65	66.47 ± 21.82	61.90 ± 20.30	69.00 ± 69.20	68.73 ± 61.85	68.97 ± 62.09	68.27 ± 68.30	63.17 ± 63.38	60.83 ± 60.31	67.21 ± 65.74	0.18									

PEMS08

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	2tta-ols	2tta-staff
MAE	3	63.20 ± 1.13	43.94 ± 0.90	40.82 ± 1.15	45.12 ± 0.66	46.25 ± 2.35	42.52 ± 0.86	43.03 ± 0.30	42.97 ± 0.15	45.08 ± 0.35	45.58 ± 0.43	44.37 ± 0.38	44.43 ± 0.93	43.52 ± 0.10	40.42 ± 0.37	47.00 ± 0.13	43.76 ± 0.40	44.06 ± 0.22	42.36 ± 0.11	41.68 ± 0.05	
	6	63.50 ± 1.45	45.34 ± 0.91	41.03 ± 1.59	46.02 ± 0.24	44.61 ± 0.13	43.94 ± 0.44	44.16 ± 0.14	44.08 ± 0.06	46.04 ± 0.44	45.58 ± 0.15	45.21 ± 0.39	43.83 ± 0.94	44.61 ± 0.10	41.70 ± 0.28	48.02 ± 0.44	44.86 ± 0.15	45.61 ± 0.13	43.50 ± 0.12	42.35 ± 0.08	
	12	64.57 ± 1.48	48.35 ± 0.91	41.63 ± 1.82	45.80 ± 0.16	46.83 ± 0.15	45.90 ± 0.16	46.23 ± 0.16	47.86 ± 0.16	46.59 ± 0.16	46.86 ± 0.43	45.52 ± 0.96	43.83 ± 0.10	40.84 ± 0.20	40.39 ± 0.17	47.03 ± 0.18	48.95 ± 0.15	45.52 ± 0.13	43.43 ± 0.14		
	Avg	63.68 ± 1.45	45.61 ± 0.91	41.11 ± 1.62	46.03 ± 0.24	44.79 ± 0.14	44.05 ± 0.14	44.28 ± 0.14	44.20 ± 0.16	46.15 ± 0.44	44.73 ± 0.15	45.31 ± 0.40	43.32 ± 0.94	44.79 ± 0.10	41.83 ± 0.28	48.28 ± 0.15	45.02 ± 0.15	45.91 ± 0.13	43.60 ± 0.12	42.37 ± 0.09	
RMSE	3	81.62 ± 1.72	42.93 ± 0.38	38.77 ± 0.23	39.91 ± 0.65	42.14 ± 0.22	41.30 ± 0.21	41.54 ± 0.21	41.43 ± 0.24	44.01 ± 0.22	42.12 ± 0.21	42.77 ± 0.52	42.13 ± 0.12	42.14 ± 0.70	40.06 ± 0.25	45.96 ± 0.22	44.46 ± 0.23	43.19 ± 0.20	40.80 ± 0.19	49.62 ± 0.08	
	6	82.29 ± 1.75	45.57 ± 0.39	39.03 ± 0.25	42.67 ± 0.66	44.36 ± 0.23	43.36 ± 0.23	43.76 ± 0.23	43.64 ± 0.25	45.96 ± 0.24	44.19 ± 0.24	44.47 ± 0.55	45.04 ± 0.12	44.36 ± 0.70	40.32 ± 0.27	47.68 ± 0.24	44.59 ± 0.26	46.15 ± 0.23	43.00 ± 0.20	40.98 ± 0.18	
	12	84.13 ± 1.80	48.71 ± 0.39	39.63 ± 0.29	42.71 ± 0.67	44.30 ± 0.28	44.47 ± 0.26	46.88 ± 0.27	47.55 ± 0.27	47.39 ± 0.29	47.99 ± 0.27	47.43 ± 0.61	44.94 ± 0.12	48.47 ± 0.70	40.24 ± 0.21	47.57 ± 0.28	48.49 ± 0.31	46.98 ± 0.26	46.64 ± 0.23	43.08 ± 0.38	
	Avg	82.53 ± 1.75	45.93 ± 0.39	39.08 ± 0.26	42.08 ± 0.66	44.31 ± 0.24	43.51 ± 0.23	43.92 ± 0.23	43.80 ± 0.26	46.15 ± 0.24	44.41 ± 0.24	44.59 ± 0.55	45.82 ± 0.12	44.61 ± 0.70	40.39 ± 0.28	48.08 ± 0.24	44.82 ± 0.26	46.58 ± 0.23	43.12 ± 0.21	41.01 ± 0.21	
MAPE (%)	3	316.71 ± 8.49	29.29 ± 0.58	18.18 ± 0.21	16.67 ± 0.58	28.26 ± 0.20	17.17 ± 0.18	18.65 ± 0.18	18.36 ± 0.19	22.24 ± 0.22	20.52 ± 0.20	20.80 ± 0.26	26.09 ± 0.16	20.17 ± 0.13	13.03 ± 0.36	16.11 ± 0.21	14.46 ± 0.18	18.99 ± 0.16	16.95 ± 0.16	16.69 ± 0.16	
	6	317.48 ± 8.49	29.71 ± 0.58	18.58 ± 0.22	17.51 ± 0.57	27.75 ± 0.21	17.23 ± 0.19	19.77 ± 0.19	19.38 ± 0.20	23.33 ± 0.23	21.54 ± 0.21	21.77 ± 0.26	26.29 ± 0.19	21.45 ± 0.12	11.23 ± 0.53	17.52 ± 0.21	17.55 ± 0.20	16.65 ± 0.17	17.91 ± 0.17	17.13 ± 0.19	
	12	318.88 ± 8.48	28.18 ± 0.60	19.00 ± 0.25	17.09 ± 0.57	27.48 ± 0.23	17.36 ± 0.22	18.86 ± 0.21	18.85 ± 0.25	25.75 ± 0.23	23.98 ± 0.27	27.74 ± 0.26	26.63 ± 0.12	23.76 ± 0.54	11.62 ± 0.41	17.10 ± 0.25	17.07 ± 0.24	18.82 ± 0.20	18.11 ± 0.18	18.09 ± 0.23	
	Avg	317.57 ± 8.48	28.08 ± 0.58	18.82 ± 0.22	17.87 ± 0.57	27.92 ± 0.21	17.50 ± 0.20	19.66 ± 0.19	19.59 ± 0.20	23.74 ± 0.23	22.00 ± 0.26	25.54 ± 0.60	26.35 ± 0.12	21.50 ± 0.53	13.59 ± 0.37	17.96 ± 0.22	17.83 ± 0.21	18.14 ± 0.18	18.15 ± 0.17	17.23 ± 0.18	

PEMS10

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	2tta-ols	2tta-staff
MAE	3	10.50 ± 0.09	9.95 ± 0.01	10.98 ± 0.10	10.41 ± 0.03	3.95 ± 0.39	1.69 ± 0.09	0.39 ± 0.09	0.56 ± 0.09	0.70 ± 0.01	0.45 ± 0.19	0.73 ± 0.01	0.59 ± 0.25	0.66 ± 0.39	0.69 ± 0.01	0.43 ± 0.11	0.42 ± 0.09	0.78 ± 0.01	0.10 ± 0.09	0.18 ± 0.08	0.59 ± 0.05
	6	10.74 ± 0.10	10.69 ± 0.11	11.37 ± 0.10	10.85 ± 0.04	4.23 ± 0.10	1.25 ± 0.09	0.90 ± 0.01	0.13 ± 0.10	0.26 ± 0.00	0.96 ± 0.00	0.26 ± 0.12	0.13 ± 0.25	0.94 ± 0.10	0.25 ± 0.00	0.90 ± 0.11	0.84 ± 0.10	0.32 ± 0.00	0.92 ± 0.09	0.72 ± 0.08	0.80 ± 0.05
	12	11.74 ± 0.12	12.23 ± 0.12	12.38 ± 0.12	12.01 ± 0.04	4.43 ± 0.11	1.43 ± 0.10	0.92 ± 0.11	0.26 ± 0.11	0.44 ± 0.12	0.05 ± 0.13	0.38 ± 0.13	0.42 ± 0.29	0.38 ± 0.11	0.43 ± 0.12	0.12 ± 0.12	0.88 ± 0.11	0.44 ± 0.11	0.71 ± 0.10	0.76 ± 0.10	0.16 ± 0.05
	Avg	10.92 ± 0.10	10.82 ± 0.11	11.50 ± 0.10	10.99 ± 0.04	4.20 ± 0.10	1.36 ± 0.09	0.98 ± 0.01	0.22 ± 0.10	0.37 ± 0.10	0.06 ± 0.10	0.36 ± 0.12	0.27 ± 0.26	0.73 ± 0.10	0.36 ± 0.10	0.04 ± 0.11	0.96 ± 0.11	0.41 ± 0.10	0.09 ± 0.09	0.79 ± 0.08	0.82 ± 0.05
RMSE	3	16.84 ± 0.16	16.00 ± 0.19	19.16 ± 0.16	16.54 ± 0.06	6.11 ± 0.15	3.46 ± 0.05	1.03 ± 0.15	0.23 ± 0.15	0.47 ± 0.17	0.13 ± 0.13	0.54 ± 0.25	0.21 ± 0.17	0.95 ± 0.35	0.46 ± 0.15	0.56 ± 0.16	0.70 ± 0.17	0.65 ± 0.16	0.19 ± 0.14	0.72 ± 0.13	0.75 ± 0.09
	6	17.43 ± 0.17	17.37 ± 0.19	19.87 ± 0.17	17.41 ± 0.06	6.77 ± 0.16	3.58 ± 0.06	1.03 ± 0.16	0.36 ± 0.16	0.57 ± 0.18	0.14 ± 0.16	0.61 ± 0.26	0.26 ± 0.37	0.22 ± 0.16	0.58 ± 0.17	0.44 ± 0.18	0.45 ± 0.16	0.71 ± 0.17	0.71 ± 0.15	0.77 ± 0.14	0.19 ± 0.09
	12	19.30 ± 0.20	20.05 ± 0.21	21.60 ± 0.19	19.46 ± 0.07	7.72 ± 0.18	3.82 ± 0.17	0.92 ± 0.18	0.53 ± 0.18	0.75 ± 0.20	0.14 ± 0.18	0.74 ± 0.20	0.43 ± 0.43	0.59 ± 0.15	0.82 ± 0.19	0.48 ± 0.20	0.22 ± 0.18	0.80 ± 0.20	0.80 ± 0.17	0.77 ± 0.14	0.90 ± 0.08
	Avg	17.71 ± 0.17	17.57 ± 0.20	20.07 ± 0.17	17.63 ± 0.06	6.80 ± 0.16	3.76 ± 0.16	0.16 ± 0.16	0.51 ± 0.16	0.75 ± 0.18	0.30 ± 0.16	0.78 ± 0.26	0.43 ± 0.38	0.41 ± 0.16	0.76 ± 0.18	0.65 ± 0.18	0.64 ± 0.16	0.87 ± 0.17	0.97 ± 0.15	0.91 ± 0.14	0.21 ± 0.08
MAPE (%)	3	32.35 ± 0.29	29.97 ± 0.31	41.64 ± 0.31	31.72 ± 0.11	18.50 ± 0.34	3.31 ± 0.29	0.29 ± 0.30	0.84 ± 0.32	0.37 ± 0.31	0.02 ± 0.11	0.31 ± 0.13	0.67 ± 0.66	0.78 ± 0.18	0.67 ± 0.35	0.31 ± 0.14	0.44 ± 0.20	0.94 ± 0.10	0.48 ± 0.10	0.57 ± 0.28	0.49 ± 0.13
	6	33.04 ± 0.31	31.54 ± 0.32	42.45 ± 0.32	32.65 ± 0.11	16.34 ± 0.32	3.39 ± 0.30	0.34 ± 0.31	0.69 ± 0.33	0.62 ± 0.32	0.01 ± 0.12	0.39 ± 0.12	0.39 ± 0.67	0.51 ± 0.15	0.39 ± 0.32	0.94 ± 0.15	0.66 ± 0.21	0.97 ± 0.11	0.50 ± 0.12	0.35 ± 0.29	0.90 ± 0.17
	12	35.48 ± 0.35	35.40 ± 0.35	45.14 ± 0.35	35.43 ± 0.11	11.79 ± 0.29	3.20 ± 0.29	0.87 ± 0.34	0.31 ± 0.37	0.22 ± 0.34	0.78 ± 0.15	0.01 ± 0.11	0.78 ± 0.71	0.50 ± 0.13	0.20 ± 0.33	0.44 ± 0.19	0.96 ± 0.24	0.53 ± 0.17	0.55 ± 0.31	0.31 ± 0.28	0.89 ± 0.21
	Avg	33.42 ± 0.31	31.97 ± 0.32	42.85 ± 0.33	33.04 ± 0.11	16.03 ± 0.32	3.75 ± 0.30	0.64 ± 0.32	0.07 ± 0.34	0.11 ± 0.32	0.38 ± 0.12	0.71 ± 0.12	0.33 ± 0.68	0.18 ± 0.13	0.75 ± 0.32	0.39 ± 0.16	0.09 ± 0.22	0.26 ± 0.13	0.09 ± 0.13	0.58 ± 0.29	0.01 ± 0.16

PEMS11

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	2tta-ol	2tta-sta	
MAE	3	52.60 ± 9.6	49.96 ± 8.4	42.02 ± 5.8	47.47 ± 8.7	20.41 ± 15.91	20.97 ± 16.27	24.70 ± 37.02	23.34 ± 56.94	106.41 ± 0.6	42 ± 20	16 ± 0.6	66 ± 0.7	22 ± 0.4	64 ± 13	95 ± 0.						
	6	53.24 ± 9.8	49.93 ± 8.4	44.14 ± 5.9	45.65 ± 8.8	9.09 ± 0.7	0.752 ± 0.769	0.802 ± 26.11	38.73 ± 25.02	98.28 ± 108.09	0.741 ± 21.69	0.832 ± 0.930	0.638 ± 14.97	0.								
	12	55.51 ± 9.1	43.17 ± 8.4	48.89 ± 5.2	42.59 ± 9.2	21.58 ± 20.73	20.99 ± 21.48	29.03 ± 22.20	29.10 ± 42.94	21.58 ± 20.16	24.84 ± 21.85	23.94 ± 0.971	16.88 ± 0.									
	Avg	53.61 ± 9.9	49.32 ± 8.4	42.29 ± 5.0	45.25 ± 8.9	18.39 ± 0.77	0.792 ± 0.829	26.35 ± 39.02	25.48 ± 59.09	98.39 ± 0.77	21.98 ± 0.864	0.976 ± 0.661	15.09 ± 0.									
RMSE	3	70.68 ± 126	86 ± 131	75 ± 78	77 ± 110	71 ± 25	77 ± 25	09 ± 25	19 ± 25	47 ± 39	42 ± 26	42 ± 34	80 ± 75	77 ± 125	77 ± 25	10 ± 30	89 ± 26	02 ± 2.7	31 ± 2.3	47 ± 22	29 ± 0.	
	6	72.15 ± 130	73 ± 131	90 ± 131	13 ± 51	37 ± 113	75 ± 29	10 ± 28	24 ± 28	48 ± 28	86 ± 41	74 ± 59	83 ± 37	96 ± 58	80 ± 129	10 ± 26	91 ± 43	45 ± 29	27 ± 3.1	45 ± 26	83 ± 24	42 ± 0.
	12	76.41 ± 138	49 ± 132	76 ± 132	74 ± 120	15 ± 25	53 ± 34	05 ± 34	47 ± 35	11 ± 46	47 ± 46	12 ± 46	86 ± 57	00 ± 135	53 ± 31	66 ± 48	62 ± 35	68 ± 39	91 ± 32	86 ± 28	10 ± 28	0 ± 0.
	Avg	72.73 ± 131	73 ± 132	86 ± 132	71 ± 114	71 ± 29	55 ± 28	60 ± 28	83 ± 29	25 ± 42	11 ± 50	22 ± 38	61 ± 79	96 ± 129	55 ± 27	48 ± 43	87 ± 29	75 ± 32	15 ± 2.7	15 ± 2.7	57 ± 24	57 ± 0.
MAPE (%)	3	248.33 ± 60	84 ± 51	33 ± 23	62 ± 44	74.74 ± 122.70	70.21 ± 21.91	21.21 ± 23.02	32.36 ± 25.09	39.73 ± 72.95	122.70 ± 27.97	26.52 ± 23.88	21.85 ± 0.979	18.93 ± 0.								
	6	249.14 ± 60	93 ± 51	81 ± 24	86 ± 17	69 ± 12	48 ± 23	34 ± 22	79 ± 24	55 ± 33	87 ± 26	49 ± 41	50 ± 94	59 ± 12	48 ± 29	14 ± 28	85 ± 25	44 ± 24	04 ± 21	24 ± 19	74 ± 0.	
	12	251.63 ± 59	83 ± 53	62 ± 28	27 ± 84	80 ± 12	22 ± 26	72 ± 26	37 ± 28	36 ± 60	44 ± 60	18 ± 46	71 ± 91	27 ± 128	22 ± 33	05 ± 23	64 ± 28	98 ± 29	63 ± 24	30 ± 21	36 ± 0.	
	Avg	249.45 ± 60	84 ± 52	89 ± 25	29 ± 18	58 ± 12	78 ± 23	70 ± 23	16 ± 25	02 ± 34	25 ± 26	96 ± 42	20 ± 75	82 ± 12	48 ± 29	72 ± 29	27 ± 25	80 ± 24	72 ± 21	52 ± 19	86 ± 0.	

PEMS12

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	2tta-ols	2tta-staff
MAE	3	30.52 ± 1.30	31.11 ± 1.30	30.59 ± 1.43	33.64 ± 1.04	28.82 ± 1.25	25.54 ± 1.22	22.24 ± 1.22	22.26 ± 1.22	22.24 ± 1.38	23.81 ± 1.22	22.21 ± 1.28	22.88 ± 1.32	32.48 ± 1.22	22.54 ± 1.33	23.43 ± 1.25	25.21 ± 1.27	22.71 ± 1.31	23.12 ± 1.10	21.08 ± 1.38	20.38 ± 0.06
	6	30.89 ± 1.44	34.35 ± 1.01	31.05 ± 1.44	44.47 ± 1.57	25.79 ± 1.35	23.55 ± 1.31	23.14 ± 1.31	23.31 ± 1.32	23.29 ± 1.47	23.23 ± 1.36	23.65 ± 1.33	23.02 ± 1.35	23.55 ± 1.41	24.19 ± 1.20	26.20 ± 1.37	23.74 ± 1.44	24.49 ± 1.21	22.17 ± 1.09	20.96 ± 0.06	
	12	32.40 ± 1.17	37.13 ± 1.21	32.56 ± 1.46	46.49 ± 1.76	25.68 ± 1.56	23.64 ± 1.48	23.82 ± 1.53	23.31 ± 1.53	23.30 ± 1.66	23.19 ± 1.55	23.29 ± 1.55	23.00 ± 1.55	23.64 ± 1.62	26.29 ± 1.86	28.64 ± 1.59	23.94 ± 1.74	24.49 ± 1.42	24.20 ± 1.11	21.84 ± 0.07	
	Avg	31.14 ± 1.44	34.62 ± 1.18	31.28 ± 1.44	44.69 ± 1.59	25.93 ± 1.37	23.72 ± 1.32	23.24 ± 1.34	23.44 ± 1.34	23.42 ± 1.48	23.36 ± 1.37	23.79 ± 1.33	23.29 ± 1.37	23.72 ± 1.45	24.45 ± 1.64	26.49 ± 1.39	23.94 ± 1.77	24.29 ± 1.22	20.97 ± 0.06		
RMSE	3	45.00 ± 1.22	47.80 ± 1.55	45.46 ± 1.23	53.14 ± 1.03	39.98 ± 1.21	31.62 ± 1.31	21.25 ± 1.25	21.32 ± 1.23	23.70 ± 1.32	21.32 ± 1.21	21.96 ± 1.47	27.90 ± 1.21	21.62 ± 1.23	23.23 ± 1.25	25.07 ± 1.21	21.90 ± 1.22	22.93 ± 1.20	20.11 ± 1.09	19.07 ± 0.06	
	6	45.98 ± 1.25	54.43 ± 1.56	46.48 ± 1.24	54.92 ± 1.06	36.05 ± 1.23	33.80 ± 1.31	23.43 ± 1.23	23.55 ± 1.25	25.65 ± 1.23	23.47 ± 1.23	23.67 ± 1.49	29.64 ± 1.23	23.80 ± 1.24	24.79 ± 1.26	26.94 ± 1.24	24.04 ± 1.25	25.89 ± 1.22	22.32 ± 1.20	20.42 ± 0.09	
	12	48.98 ± 1.30	60.74 ± 1.59	49.28 ± 1.28	58.72 ± 1.09	39.75 ± 1.27	37.83 ± 1.26	26.41 ± 1.27	27.27 ± 1.27	29.31 ± 1.27	27.21 ± 1.26	26.75 ± 1.53	33.94 ± 1.27	27.83 ± 1.28	28.54 ± 1.31	31.31 ± 1.28	28.21 ± 1.31	31.74 ± 1.26	26.08 ± 1.22	22.27 ± 0.15	
	Avg	46.38 ± 1.25	54.84 ± 1.56	46.84 ± 1.25	55.25 ± 1.06	36.24 ± 1.24	34.04 ± 1.32	23.32 ± 1.23	23.62 ± 1.23	23.74 ± 1.25	23.64 ± 1.23	23.83 ± 1.50	30.01 ± 1.24	24.04 ± 1.25	25.17 ± 1.27	27.41 ± 1.24	24.33 ± 1.26	26.33 ± 1.22	22.47 ± 1.20	20.38 ± 0.10	
MAPE (%)	3	189.12 ± 21.22	214.22 ± 135.89	244.80 ± 27.93	255.12 ± 23.97	222.82 ± 23.83	227.15 ± 24.21	228.87 ± 28.94	225.12 ± 25.12	229.91 ± 24.93	226.55 ± 23.91	221.05 ± 20.92	20.14								
	6	189.83 ± 25.25	215.06 ± 26.07	228.97 ± 26.28	244.98 ± 23.96	224.78 ± 23.11	225.43 ± 29.43	220.19 ± 26.28	221.55 ± 22.52	227.55 ± 26.21	222.42 ± 21.59	20.15									
	12	192.38 ± 31.60	219.47 ± 29.76	231.53 ± 29.32	227.65 ± 27.15	227.91 ± 30.63	228.60 ± 31.89	220.43 ± 29.32	226.84 ± 20.18	220.38 ± 22.33	225.40 ± 22.96	20.17									
	Avg	190.26 ± 25.91	216.87 ± 26.58	229.25 ± 26.64	225.29 ± 24.35	225.20 ± 23.38	225.81 ± 29.83	220.49 ± 26.64	222.31 ± 23.41	227.94 ± 26.98	222.71 ± 21.70	20.17									

TFNSW

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	2tta-ols	2tta-staff
MAE	3	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	72.94 ± 63.76 ± 0.73	72.94 ± 63.76 ± 0.73
	6	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	87.17 ± 70.62 ± 0.71	87.17 ± 70.62 ± 0.71
	12	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	98.55 ± 78.81 ± 0.57	98.55 ± 78.81 ± 0.57
	Avg	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	84.53 ± 70.05 ± 0.66	84.53 ± 70.05 ± 0.66
RMSE	3	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	153.55 ± 144.09 ± 0.7	153.55 ± 144.09 ± 0.7
	6	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	182.81 ± 161.63 ± 0.4	182.81 ± 161.63 ± 0.4
	12	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	209.14 ± 181.83 ± 1.1	209.14 ± 181.83 ± 1.1
	Avg	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	177.14 ± 159.68 ± 0.5	177.14 ± 159.68 ± 0.5
MAPE (%)	3	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	66.03 ± 54.94 ± 1.21	66.03 ± 54.94 ± 1.21
	6	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	82.83 ± 60.17 ± 1.18	82.83 ± 60.17 ± 1.18
	12	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	96.55 ± 69.60 ± 1.48	96.55 ± 69.60 ± 1.48
	Avg	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	81.77 ± 61.07 ± 1.13	81.77 ± 61.07 ± 1.13